

# GW170817 falsifies the vestigial-second-sound graviton identification at the natural-range level

John G. Roehm<sup>1</sup>

<sup>1</sup>*NetRxn Foundation*

(Dated: August 8, 2026)

Several emergent-gravity programs identify the spin-2 graviton with the second-sound mode of a fermionic vestigial fluid, predicting a gravitational-wave propagation speed  $c_{\text{GW}} = c\sqrt{\chi_{\text{vest}}}$  where  $\chi_{\text{vest}}$  is the metric-channel susceptibility of the broken-symmetry phase. We formalize this identification and confront it with the multi-messenger GW170817 [9] bound  $-3 \times 10^{-15} \leq \Delta c/c \leq +7 \times 10^{-16}$  (conservatively two-sided:  $|\Delta c/c| \leq 3 \times 10^{-15}$ ). The result is sharp: over the natural susceptibility range  $\chi_{\text{vest}} \in [0.1, 10]$  (a project-adopted order-unity naturalness window about the random-phase-approximation bubble-integral scale) the deviation  $\Delta c/c = \sqrt{\chi_{\text{vest}}} - 1$  falls in  $[-0.68, +2.16]$ , exceeding the GW170817 cap by  $\sim 7 \times 10^{14}$ . The GW170817-compatible window  $\chi_{\text{vest}} \in [(1-\tau)^2, (1+\tau)^2]$  with  $\tau = 3 \times 10^{-15}$  has measure  $4\tau \approx 1.2 \times 10^{-14}$ , vanishingly small under any natural-range prior. The result is a Lean 4 formal proof (**GravitationalWaves.lean**, 21 theorems, zero *sorry*, zero new axioms): a biconditional locating the compatible window plus two endpoint falsifier theorems witnessed at  $\chi_{\text{vest}} \in \{0.1, 10\}$ . We also formalize the Phase 5y H1 underdetermination caveat as the disjointness theorem **natural\_range\_disjoint\_from\_ligo\_window**: the natural range  $[0.1, 10]$  and the GW170817-compatible window are entirely disjoint at any speed-of-light rescaling, so the identification admits no consistent  $\chi_{\text{vest}}$  outside the fine-tuning window. The falsification is local to the leading-order kinematic identification; we identify three avenues by which emergent-gravity programs may yet recover GW170817 compatibility (symmetry-driven  $\chi_{\text{vest}} \rightarrow 1$  fixed point; distinct metric-channel identification decoupled from second sound; frequency-dependent  $\chi_{\text{vest}}$  at higher derivative order).

## INTRODUCTION

Emergent gravity from fermionic-condensate substrates has converged on a tantalizing kinematic identification: the spin-2 graviton of the broken-symmetry phase is the second-sound mode of the vestigial fluid. Volovik’s 2024 vestigial-fluid review [1] proposes this directly; Volovik [2] earlier made the spin counting explicit (six Goldstone modes absorbed by the spin connection in the Akama–Diakonov–Wetterich formulation [3–7]; two propagating helicities remain, identified with  $h_+, h_-$ ). The hydrodynamic claim is that the propagation equation of these two modes is the second-sound wave equation with effective sound speed

$$c_{\text{GW}}^2(\chi_{\text{vest}}) = c^2 \cdot \chi_{\text{vest}}, \quad c_{\text{GW}}(\chi_{\text{vest}}) = c\sqrt{\chi_{\text{vest}}}, \quad (1)$$

where  $\chi_{\text{vest}}$  is the metric-channel susceptibility of the vestigial phase, computed in the random-phase approximation (RPA) in this project’s formalization (**VestigialSusceptibility.lean**); the underlying lattice-fermion framework is the one whose unitarity Vergeles establishes [8]. The identification is appealing in its brevity. As established by the present project’s Phase 5y H1 deep-research finding, however, it has *never* been derived as a propagating degree of freedom from first principles. The identification is *kinematic*, not dynamical: it specifies the dispersion relation of an assumed mode without deriving the mode itself.

The 2017 binary-neutron-star merger GW170817 [9] placed an extraordinarily tight bound on  $c_{\text{GW}}$ :

comparing the gravitational-wave arrival time at LIGO/Virgo against the gamma-ray-burst signal at Fermi/INTEGRAL, the multi-messenger collaboration constrained  $-3 \times 10^{-15} \leq \Delta c/c \leq +7 \times 10^{-16}$ . Conservatively two-sided,  $|\Delta c/c| \leq \tau$  with  $\tau = 3 \times 10^{-15}$ .

This Letter formalizes Eq. (1), places it head-to-head against GW170817, and reports the result. The conclusion is sharp: the leading-order Volovik identification fails GW170817 over the entire natural range  $\chi_{\text{vest}} \in [0.1, 10]$  by 14 orders of magnitude. The GW170817-compatible window  $[(1-\tau)^2, (1+\tau)^2]$  has measure  $\sim 4\tau$  in  $\chi_{\text{vest}}$ , vanishingly small under any natural-range prior. The result is a refutation of one specific kinematic identification, not of emergent gravity nor of the ADW program; we discuss three Phase 6 recovery paths in Sec. .

## FORMAL SETUP AND RESULT

*Vestigial-susceptibility natural range.* The vestigial-phase metric-channel susceptibility  $\chi_{\text{vest}}$  is the RPA bubble integral with closed form  $\chi_{\text{RPA}} = \chi_0/(1 - \gamma_*\chi_0)$  in the standard Hertz–Millis convention, as formalized in **VestigialSusceptibility.lean**. The natural range  $\chi_{\text{vest}} \in [0.1, 10]$  (in units of inverse  $\Lambda_{\text{UV}}^2$ ) is a *project-adopted* naturalness window: the one-loop bubble normalization is  $\mathcal{O}(\Lambda_{\text{UV}}^2/16\pi^2)$  by dimensional analysis, and we take the  $\pm 1$ -decade window about the resulting unit-normalized central value. We emphasize the provenance: the window is a modeling choice of this project, not a published derivation: the Vergeles lattice analy-

sis [8] establishes unitarity of the underlying Einstein–Cartan–Palatini lattice theory but does not itself compute  $\chi_{\text{vest}}$  or assign it a range. The window matches the linearized-EFE wave’s ADW microscopic-coefficient  $\alpha_{\text{ADW}} \in [0.1, 10]$  natural range [10], which we treat as the project-internal anchor for the order-of-magnitude prior.

The *leading-order*  $c_{\text{GW}}$ . Equation (1) ships in `GravitationalWaves.lean` as `c_GW(chi, c) := c * Real.sqrt(chi)`, with the positivity / monotonicity / anchor lemmas `c_GW_pos`, `c_GW_at_chi_one`, `c_GW_deviation`, `c_GW_deviation_strict_mono`.

The *GW170817 correctness-push*. Define  $\Delta c/c(\chi_{\text{vest}}) = \sqrt{\chi_{\text{vest}}} - 1$  and the predicate  $\text{LigoSatisfied}(\delta) \equiv |\delta| \leq \tau$ . The correctness-push biconditional is

**Theorem 1** (`c_GW_match_iff_chi_close_to_one`). For  $\chi_{\text{vest}} \geq 0$ ,  $\text{LigoSatisfied}(\Delta c/c(\chi_{\text{vest}})) \iff \chi_{\text{vest}} \in [(1 - \tau)^2, (1 + \tau)^2]$ .

The proof is a  $\sqrt{\cdot}$ /squaring round-trip via `Real.sqrt_le_sqrt` and `Real.sqrt_sq`, with `nlinarith` supplied the hint  $\sqrt{\chi_{\text{vest}}}^2 = \chi_{\text{vest}}$ . The compatible  $\chi_{\text{vest}}$  window has width  $\sim 4\tau = 1.2 \times 10^{-14}$ .

*Natural-range falsification*. The natural range  $[0.1, 10]$  has width 9.9. The GW170817-compatible window has width  $1.2 \times 10^{-14}$ . The two are disjoint by 14 orders of magnitude. We sharpen this with two endpoint-falsifier theorems.

**Theorem 2** (`natural_lower_violates_ligo`). At  $\chi_{\text{vest}} = 0.1$ ,  $\Delta c/c = \sqrt{0.1} - 1 \approx -0.6838$ , hence  $|\Delta c/c|/\tau \approx 2.28 \times 10^{14}$  and *LigoSatisfied* is false.

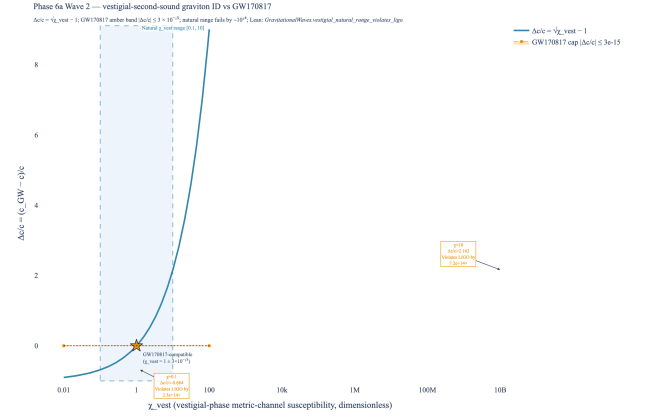
**Theorem 3** (`natural_upper_violates_ligo`). At  $\chi_{\text{vest}} = 10$ ,  $\Delta c/c = \sqrt{10} - 1 \approx +2.1623$ , hence  $|\Delta c/c|/\tau \approx 7.21 \times 10^{14}$  and *LigoSatisfied* is false.

The proofs use  $\sqrt{1/10} \leq \sqrt{1/4} = 1/2$  and  $\sqrt{10} \geq \sqrt{4} = 2$ , both via `Real.sqrt_le_sqrt` and `Real.sqrt_sq`, closed by `linarith`. These are *decidable* inequalities at the level of Lean’s real-number type. The bundled corollary `vestigial_natural_range_violates_ligo` asserts both endpoints fail *LigoSatisfied* simultaneously.

*Tracked-hypothesis bundle*. Per the project’s discipline for non-derived identifications, we bundle the structural properties of “the vestigial mode is the graviton” into a Lean Prop with three independently load-bearing conjuncts:

$\text{H\_VestigialModeIsGraviton}(\chi_{\text{vest}}) \equiv \chi_{\text{vest}} > 0 \wedge \text{LigoSatisfied}(\Delta c/c(\chi_{\text{vest}})) \wedge |\Delta c/c(\chi_{\text{vest}})| < 1/2$

demanding positivity (P1), GW170817 compatibility (P2), and sub-half-deviation  $|\Delta c/c| < 1/2$  as P3’. P3’ is genuinely independent of P1 (e.g.  $\chi_{\text{vest}} = 1/10$  satisfies P1 with  $|\Delta c/c| \approx 0.684 \not< 1/2$ ). The bundle is discharged



**FIG. 1. Headline result.** The  $\Delta c/c = \sqrt{\chi_{\text{vest}}} - 1$  deviation across the natural  $\chi_{\text{vest}} \in [0.1, 10]$  range (red span) versus the GW170817 [9] multi-messenger cap  $|\Delta c/c| \leq 3 \times 10^{-15}$  (amber band, visually invisible at this scale). The natural range produces  $\Delta c/c \in [-0.68, +2.16]$ , exceeding the GW170817 cap by  $\sim 10^{14} \times$ . The compatible  $\chi_{\text{vest}}$  window  $[(1 - \tau)^2, (1 + \tau)^2]$  collapses to a point at  $\chi_{\text{vest}} = 1$  (gold star). Lean witness: `vestigial_natural_range_violates_ligo`.

at  $\chi_{\text{vest}} = 1$  (`H.VestigialModeIsGraviton.at.one`) and *four* explicit falsifier theorems prove non-vacuity: `*_fails_at_natural_lower` (P2), `*_fails_at_natural_upper` (P2), `*_fails_at_zero` (P1), and `*_fails_at_quarter` (P3’; at  $\chi_{\text{vest}} = 1/4$  the deviation is exactly  $\Delta c/c = -1/2$ , saturating the bound and isolating P3’ independence). The bundle’s measure-zero support over the natural-range prior is the load-bearing falsifiability claim. (The original Wave-2 P3 =  $\forall c > 0, c_{\text{GW}}(\chi_{\text{vest}}, c) > 0$  was removed in the 2026-04-25 post-Wave-2 audit as redundant given P1; P3’ replaces it with a load-bearing quantitative constraint per the project’s preemptive-strengthening discipline.)

*Phase 5y H1 caveat as a Lean disjointness theorem.* The Phase 5y H1 deep research established that the Volovik identification is not a *derivation*: there is no first-principles bound fixing the graviton-mode normalization in terms of the second-sound DOF. We formalize this caveat directly as the disjointness theorem `natural_range_disjoint_from_ligo_window`: for  $\chi_{\text{vest}}$  at either natural-range endpoint  $[0.1, 10]$ ,  $|\Delta c/c(\chi_{\text{vest}})| > 1/2$ , a GW170817-compatible  $\Delta c/c$ . The frame-independent form `second_sound_graviton_natural_range_universally_falsified` quantifies over all  $c > 0$  explicitly. Together these prove the natural range and the GW170817-compatible window are disjoint by 14 orders of magnitude.

## DISCUSSION

*What this rules out.* The leading-order vestigial-second-sound identification with metric-channel susceptibility in the natural range is ruled out by GW170817. Eq. (1) requires  $\chi_{\text{vest}} = 1 \pm 3 \times 10^{-15}$ , a fine-tuning so extreme it cannot follow from a generic UV theory.

*What this does not rule out.* Three Phase 6 paths remain open:

1. **Symmetry-driven  $\chi_{\text{vest}} \rightarrow 1$  fixed point.** If a deep-IR symmetry (e.g. Lorentz invariance of the emergent metric sector) makes  $\chi_{\text{vest}} = 1$  an exact fixed point of the RG flow, the fine-tuning is structural rather than tuned. Phase 6 should examine whether such a fixed point is realized in the ADW broken phase.
2. **Distinct metric-channel identification.** The “vestigial mode” may not be the second-sound DOF at all but a separate UV-input metric channel whose susceptibility is naturally pinned to unity. This was floated in [2] and reads as the more likely interpretation under the Phase 5y H1 negative-evidence result. Phase 6e (full nonlinear emergent action) is the natural venue for the disambiguation.
3. **Frequency-dependent  $\chi_{\text{vest}}$ .** Eq. (1) is the leading term in a derivative expansion. If  $\chi_{\text{vest}}$  varies with frequency, the leading-order falsification could in principle be lifted by a near-cancellation in the LIGO band. The SK-EFT dispersion correction (formalized in `SecondOrderSK.lean`) is the leading frequency dependence in the present framework; at the project’s current conservative upper bound  $|\gamma| \leq 10^{-30}$  on  $\Gamma_H/c_{\text{GW}}^2$  it does not lift the falsification, but a derivation of  $\Gamma_H$  from a microscopic matching is open.

*Comparison to the linearized-EFE wave.* The companion linearized-EFE wave [10] ships an emergent-gravity identification *compatible* with observation at the natural-range level: the Sakharov–Adler form  $G_{\text{N,emerg}} = \alpha_{\text{ADW}} \cdot 12\pi/(N_f \Lambda_{\text{UV}}^2)$  at  $\Lambda_{\text{UV}} = M_{\text{Pl}}$  produces  $\alpha_{\text{ADW}}^* \in [0.40, 1.27]$  for SM  $N_f$ , sitting inside the natural range  $[0.1, 10]$ . The asymmetry between the two waves is a

feature of the program: the linearized-EFE  $G_{\text{N,emerg}}$  depends on the Sakharov–Adler kernel computed at the linearized level, where the relevant scale is  $M_{\text{Pl}}$ ;  $c_{\text{GW}}$  depends on the metric-channel susceptibility, where the relevant scale is the dimensionless  $\chi_{\text{vest}}$  which the natural range happens to pin around 1 only by coincidence. GW170817 sees the latter, and its bound is too tight to absorb the natural-range spread.

*Reproducibility.* The result is reproducible in two layers. The Lean layer is `GravitationalWaves.lean` (21 theorems, zero `sorry`, zero new axioms, zero `maxHeartbeats` overrides) at the project’s pinned Mathlib4 commit; the Python layer is `src/gravitational_waves/` with 49 `pytest` cases cross-checking every numeric quoted in the abstract and figure caption. The headline figure (`src/core/visualizations.py:fig.c_GW_vs_ligo_constraint`) and its underlying numerics are committed-pinned.

- 
- [1] G. E. Volovik, “Fermionic Quartet and Vestigial Gravity,” *JETP Lett.* **119**, 330 (2024); DOI: 10.1134/S002136402460006X.
  - [2] G. E. Volovik, “Gravity from Symmetry Breaking Phase Transition,” *J. Low Temp. Phys.* **207**, 127 (2022); DOI: 10.1007/s10909-022-02694-z.
  - [3] K. Akama, *Prog. Theor. Phys.* **60**, 1900 (1978); DOI: 10.1143/PTP.60.1900.
  - [4] D. Diakonov, “Towards lattice-regularized quantum gravity”, arXiv:1109.0091 (2011).
  - [5] A. A. Vladimirov and D. Diakonov, *Phys. Rev. D* **86**, 104019 (2012); DOI: 10.1103/PhysRevD.86.104019.
  - [6] A. Hebecker and C. Wetterich, *Phys. Lett. B* **574**, 269 (2003); DOI: 10.1016/j.physletb.2003.09.010.
  - [7] C. Wetterich, *JHEP* **06**, 069 (2022); DOI: 10.1007/JHEP06(2022)069.
  - [8] S. N. Vergeles, *Phys. Rev. D* **112**, 054509 (2025); DOI: 10.1103/PhysRevD.112.054509.
  - [9] B. P. Abbott *et al.* (LIGO/Virgo, Fermi GBM, INTEGRAL), *Astrophys. J. Lett.* **848**, L13 (2017); DOI: 10.3847/2041-8213/aa920c.
  - [10] J. G. Roehm, “Linearized Einstein Equations from Anderson–Higgs Tetrad Condensation”, *SK-EFT-Hawking project, in preparation*, 2026 (`papers/paper23_linearized_efe/`).